Biliary Atresia, Part Seven: Cholangitis after the Kasai Operation. Diagnosis, Empiric Therapy, Prophylaxis and Bile Lakes.

Picture a girl who is ten weeks old. Six weeks ago she had a Kasai portoenterostomy for biliary atresia, and it worked. Her stools turned yellow, her bilirubin fell, and at the last clinic visit everyone was pleased. Now her mother brings her back because she has a temperature of thirty-eight point six degrees, she is feeding less well, and the stools in her nappy are paler than they were last week. The question for the team is simple to state and surprisingly hard to answer. Is this cholangitis? If it is, which antibiotic should she get, for how long, and could it have been prevented? And what should we look for so that this does not happen again?

This chapter is about those questions. The main idea is the following. Cholangitis, a bacterial infection of the bile ducts inside the liver, is the commonest complication of a successful Kasai operation. It arrives early, often in the first two or three months, and it is usually diagnosed without any positive culture. Because fever is often missing, you should use the international definition published in 2022, which lets you diagnose cholangitis from a change in stool colour and a rise in bilirubin or gamma-glutamyl transferase, even in an infant who is not febrile. Treat it with intravenous antibiotics that cover gram-negative gut bacteria and anaerobes, think about enterococci, and continue for two to three weeks. If the child is not better after seventy-two hours, escalate. Antibiotic prophylaxis after the Kasai is used almost everywhere, but it is supported by expert consensus rather than by proof. And it is not a single episode that costs the child her liver, but early, repeated episodes, especially when they come with cystic collections of bile inside the liver, called bile lakes.

A few terms first. The Kasai portoenterostomy is the operation that removes the scarred remnant of the bile duct at the porta hepatis, the hilum of the liver, and sews a loop of small bowel, the Roux limb, onto the raw surface so that bile can drain from the tiny surviving ductules into the gut. Clearance of jaundice means that the bilirubin falls to normal or nearly normal after the operation. Native-liver survival means being alive with one's own liver, without a transplant. Gamma-glutamyl transferase, usually shortened to GGT, is a liver enzyme that rises when bile flow is obstructed. C-reactive protein, or CRP, and procalcitonin are blood markers of inflammation and bacterial infection.

The sources for this chapter include the international Delphi consensus on cholangitis after the Kasai, written by Calinescu and colleagues for the Biliary Atresia and Related Diseases network, known as BARD, and published in the Journal of Clinical Medicine in 2022; a validation of that definition from Hannover and Geneva by Madadi-Sanjani and colleagues in 2024; a review by Calinescu and Wildhaber in Seminars in Pediatric Surgery in 2024; the Hong Kong series of van Heurn and colleagues from 2003; series from Korea, China, Japan

and Israel; a multicentre Chinese randomised trial of treatment by Wang and colleagues in 2023; Taiwanese trials of prophylaxis by Bu and colleagues in 2003 and by Lien and colleagues in 2015; a Chinese randomised trial of the length of intravenous antibiotics after surgery by Chen and colleagues in 2021; meta-analyses of prophylaxis from 2016, 2023 and 2026; and series on bile lakes from Taiwan, China, Japan, Finland and Chile.

Part one. How bacteria reach the porta.

To understand cholangitis after a Kasai, start with the anatomy the surgeon creates. Normally the biliary tree is protected from the bowel by the sphincter of Oddi, a muscular valve at the end of the common bile duct. After a Kasai there is no sphincter and no valve. The Roux limb opens directly onto the porta, so the contents of the bowel can reflux back up towards the ductules. That is the first link in a chain.

The second link is stasis. In an infant whose drainage is only partial, bile moves slowly through narrow, scarred ductules. Slow bile in a narrow channel is a good culture medium. The third link is translocation. Bacteria can reach the liver not only by climbing up the Roux limb but also through the portal venous blood and the lymphatics, from a gut that is inflamed and cholestatic. The fourth link is dysbiosis, a change in the balance of the gut bacteria. In infants with biliary atresia the gut flora shifts away from protective bifidobacteria towards Klebsiella and enterococci, and antibiotic prophylaxis can select resistant strains. The fifth link is ductular loss. Each episode of cholangitis inflames and scars the remaining ductules. The bilirubin rises, and sometimes it never falls again.

This chain explains a paradox that confuses many trainees. Cholangitis needs an open channel, so infants who drain well are diagnosed with it often. In a Chinese series published in 2023, cholangitis occurred in about sixty-eight percent of infants with good early drainage and in about forty-eight percent of the others. Yet when you compare drainage groups directly, partial drainage carries the highest risk. In the Hong Kong series from 2003, infants with partial drainage had an odds ratio of about five point seven for cholangitis compared with infants who drained well. The practical lesson is that a fever in an infant whose Kasai is working is a threat to the very ductules that make the operation succeed.

Part two. How often, how early, and what it costs.

The reported frequency of cholangitis after a Kasai ranges from about one fifth of infants to more than nine in ten. Most of that spread comes from the definition rather than the disease. In the series that used broad clinical definitions, the numbers cluster between a half and four fifths. In Korea in 2020, seventy-nine percent of one hundred and sixty infants had at least one episode, and about three quarters of those had more than one. In Japan in 2024, almost seventy percent of forty-three children were affected. In China, sixty-eight percent had cholangitis within the first year in one series and sixty-six percent within two years in another. In Hong

Kong, fifty-nine percent. By contrast, the Hannover and Geneva cohort, which used the stricter consensus definition, found cholangitis in about a third of infants in the first year, and an Israeli study found it in about a quarter of infants whose Kasai had succeeded. Whichever number you choose, expect at least a third and probably half of your infants to have an episode. Most episodes occur in the first year. In one Chinese series the first episode came at a median of sixty days, and in the Japanese series almost nine in ten of all episodes happened before the age of three.

Does cholangitis cost the child her liver? The honest answer is that a single episode may not, but early and repeated episodes do. In the Hong Kong series, once the quality of drainage was taken into account, cholangitis was no longer an independent predictor of death or transplant. That is because cholangitis is both a marker and a cause: infants with partial drainage get it more and do worse anyway. Other studies separate out the episodes that matter. In the 2023 Chinese series, having any cholangitis did not change native-liver survival, but recurrent cholangitis, defined as more than three episodes in six months, clearly did. In Japan, three or more episodes predicted later transplant, although the test was only moderately accurate, with an area under the curve of about zero point seven. In a large Chinese series of infants who developed cystic lesions inside the liver, two-year native-liver survival was about forty-three percent when cholangitis began before three months of age and eighty percent when it began after six months. And in Switzerland, after a standard national protocol for cholangitis was introduced in 2015, the transplant rate fell from eighty-nine to fifty-five percent, although that is a before-and-after comparison and other things changed too. The targets, then, are prompt recognition, an adequate first course and prevention of recurrence.

Part three. Making the diagnosis.

In adults, cholangitis is diagnosed with the Tokyo guidelines, which rely on fever, abdominal pain, jaundice and signs of biliary obstruction. Infants cannot tell you about pain, and after a Kasai there is no common duct to obstruct. That is why the BARD network produced its own definition. The consensus was reached in two survey rounds of experts, followed by a working group and a panel discussion in 2021, and it was published in 2022.

The definition uses two lists. List A contains the clinical criteria. These are fever without another source, or shivering; a change in stool colour, meaning the stools become paler; new or increasing jaundice; and abdominal discomfort, which in an infant shows itself as vomiting, poor feeding or irritability. List B contains the laboratory and imaging criteria. These are a raised white cell count, CRP or procalcitonin; raised transaminases; raised GGT or bilirubin; and bile lakes seen on imaging. Suspected cholangitis needs one item from list A and one from list B. Confirmed cholangitis needs two items from each list, or suspected cholangitis together with a positive blood culture. The same group described three patterns. Early cholangitis occurs in the first month after the operation. Relapsing cholangitis means more than three

episodes within six months. And intractable cholangitis means either that antibiotics are needed for more than four weeks, or that three or more episodes occur less than a month apart.

The most important finding about this definition came from its validation in Hannover and Geneva in 2024. Among one hundred and eighty-five infants, fifty-nine had clinician-diagnosed cholangitis in the first year. When the authors looked at how often each feature was present, fever was found in only forty-one percent of episodes. Raised inflammatory markers were found in seventy percent, raised transaminases in sixty-eight percent, and a rise in bilirubin or GGT in ninety percent. Almost ninety-five percent had at least one laboratory criterion. Only one of the fifty-nine episodes had a positive blood culture, and only one showed bile lakes. The agreement between the consensus definition and the clinicians' judgement was good, with a correlation of about zero point eight. When the same group reviewed one hundred and nine published studies, fever was part of the definition in about three quarters of them, jaundice in just over half, CRP or procalcitonin in about a sixth, and a positive culture was required in about one in seven. In other words, many published definitions demand features that most real episodes do not have. In practice, paler stools and a rise in bilirubin or GGT, rather than fever, are the commonest signals, and you should act on them.

What should you do in the first hour? Take a blood culture before the first dose of antibiotic, ideally two sets. The yield is low, but a positive culture steers escalation later. Measure the full blood count, CRP and procalcitonin. In one Chinese study, procalcitonin and CRP were higher in intractable than in simple cholangitis, so they help to judge severity. Measure bilirubin, GGT and the transaminases, both as part of the definition and as a baseline for response, remembering that GGT often rises before fever appears. Do an ultrasound of the liver to look for bile lakes, ascites and portal blood flow, and to exclude another source. In selected children, culture or molecular testing of liver tissue can help: in a series from Egypt published in 2025, combining blood culture with molecular testing and liver biopsy raised the yield to about half of episodes. At the same time, think about the alternatives. Urinary infection, an infected central line, a viral illness with transient cholestasis, and, in a jaundiced infant, failure of the Kasai itself, can all look similar. A rising bilirubin after a fever should be treated as cholangitis until proved otherwise, and you should finish treating it before you conclude that the operation has failed. One small Japanese study noticed that the ratio of neutrophils to lymphocytes at the onset of an episode was lower in children who later kept their own liver than in those who needed a transplant, about zero point eight five against one point six, but that is a prognostic hint, not a decision tool.

Which bacteria are involved? When a culture is positive, enteric gram-negative bacteria and enterococci dominate. In the Korean series, enterococcus faecium made up twenty-eight percent of isolates, Escherichia coli fifteen percent, Enterobacter cloacae eleven percent and Klebsiella pneumoniae nine percent. Only about four in ten of the gram-negative isolates were

susceptible to cefotaxime, a third-generation cephalosporin, whereas nearly all were susceptible to meropenem. Among the gram-positive isolates, only about four in ten were susceptible to ampicillin, while all were susceptible to vancomycin. In the Egyptian series, Klebsiella species dominated and meropenem was most often successful. In the large Chinese treatment trial, blood cultures were positive in about fifteen percent of episodes overall and in about a quarter of severe episodes. The message is that resistance to the cephalosporins we are used to giving is rising, and that the local antibiogram and the child's own prophylactic antibiotic should shape the first choice.

Part four. Choosing and running the treatment.

There is no society guideline on which antibiotic to use, so the choice rests on consensus, on one trial and on local practice. In the BARD survey, piperacillin with tazobactam was the commonest first choice, named by about seventy-one percent of respondents, while in the published literature ceftriaxone was the commonest. The 2024 review pointed out that the efficacy of third-generation cephalosporins in these infants has fallen from about eighty-nine percent to about seventy-five percent, and that practice has moved towards piperacillin with tazobactam and then towards meropenem. A sensible frame, which is not a guideline, looks like this. For a mild first episode in a unit with little resistance, ceftriaxone or piperacillin with tazobactam is reasonable. For a moderate or severe episode, or a septic child, meropenem is the better choice because it covers organisms that produce extended-spectrum beta-lactamases. A child who develops cholangitis while taking prophylaxis should receive an antibiotic from a different class, because prophylaxis selects resistant Klebsiella. If the culture grows gram-positive cocci, or the child does not respond, add vancomycin, because enterococcus faecium led the Korean blood cultures. And if the child is not better after seventy-two hours, escalate, repeat the culture and repeat the ultrasound, looking for bile lakes, an abscess or an infected line. Always narrow the treatment when a culture result returns.

The only multicentre randomised trial of treatment was carried out in fourteen Chinese centres and published by Wang and colleagues in 2023. Episodes were graded as mild, moderate or severe on the basis of temperature, white cell count and CRP. In mild episodes, cefoperazone with sulbactam worked as well as meropenem, which means a carbapenem can be spared when the episode is mild. Moderate episodes received meropenem. In severe episodes, infants were randomised to meropenem alone or meropenem with intravenous immunoglobulin, and those who received immunoglobulin had a shorter fever, about fifty-two hours against sixty-eight hours. Secondary reports of the trial disagree about whether immunoglobulin also reduced recurrence, so that question is not settled. An earlier Chinese cohort of one hundred and thirteen infants with intractable cholangitis, published in 2018, found that immunoglobulin shortened the fever and the hospital stay and lengthened the time to the next episode. Neither study was blinded. Immunoglobulin is therefore a reasonable addition for a severe or intractable episode, but not routine care.

How long should treatment last? The BARD consensus recommends ten to fourteen days for suspected cholangitis and fourteen to twenty-one days for confirmed cholangitis, and the experts specifically judged seven to ten days to be too short. An early switch to oral antibiotics after a few days is not supported by any data in these infants. For intractable cholangitis, some centres have used six to twelve weeks of intravenous antibiotics at home, but this comes from a series of only twenty patients. There is no verified evidence that steroids treat cholangitis, and the Japanese guideline of 2021 makes no recommendation on steroids at all. When episodes run into one another, or need more than four weeks of antibiotics, the question changes. It is no longer which antibiotic to use, but whether the native liver is still worth the infections, and the transplant team should be involved early.

Part five. Can cholangitis be prevented?

Antibiotic prophylaxis after the Kasai is near-universal, yet the evidence behind it is thin. The most quoted study comes from Taiwan in 2003. Infants who had already had one episode of cholangitis were randomised to trimethoprim with sulfamethoxazole, nine infants, or to neomycin, ten infants, and both groups were compared with eighteen historical controls. Both drugs were followed by fewer recurrences than in the controls. Note two things: this was secondary prevention, after a first episode, and the comparison group was historical rather than randomised. A pilot study from the same group in 2015 compared a probiotic, Lactobacillus casei rhamnosus, with neomycin, ten infants in each group. Cholangitis occurred in one in five infants in each group, against four in five historical controls. Again the numbers are tiny and the controls historical.

Two Chinese studies looked at the antibiotics given straight after surgery. A randomised trial in 2021 assigned one hundred and eighty infants to seven or fourteen days of intravenous antibiotics after the operation. Early cholangitis was more common with the short course, sixty-one percent against thirty-eight percent, although the rate at six months and native-liver survival were similar, and the authors concluded that the shorter course was acceptable. A retrospective study from 2019 compared imipenem with a cephalosporin plus metronidazole and reported far less early cholangitis with imipenem, about ten percent against fifty-nine percent, but it was not randomised.

When the studies of oral prophylaxis are pooled, the signal disappears. A systematic review in 2016 found four studies with three hundred and twenty-nine infants and judged the evidence too weak to reach a conclusion. A meta-analysis in 2023 pooled six studies and seven hundred and fourteen infants and found no reduction in cholangitis, with an odds ratio of about one point one five and a confidence interval running from about zero point six to more than two. A further meta-analysis published in 2026 found cholangitis in about fifty-seven percent of infants with prophylaxis and fifty-seven percent without. It did report better native-liver survival with prophylaxis, an odds ratio of about two, but that estimate came from only two

studies and the authors themselves called it hypothetical. An Israeli multicentre study in 2025 found that infants on primary prophylaxis had their first episode earlier, at a median of seventy-seven days against two hundred and thirty-nine days, which raises the possibility that prophylaxis delays recognition or selects more resistant bacteria. A randomised trial is now under way in Shanghai, comparing trimethoprim with sulfamethoxazole, alternating every two weeks with cefaclor, from day fifteen to month six, against no prophylaxis, with a target of three hundred and fifty-six infants. It is not expected to report until about 2029.

Until then, the BARD consensus supports oral prophylaxis for six to twelve months after the operation, without naming a drug. Trimethoprim with sulfamethoxazole is the most used, chosen by just over half of the surveyed experts. In the Taiwanese trial the dose was trimethoprim four milligrams per kilogram per day, and neomycin was given at twenty-five milligrams per kilogram per day on four days a week. If you give prophylaxis, choose one agent, write down its stop date, and treat breakthrough cholangitis with an antibiotic from a different class. Watch for resistant Klebsiella and cytopenias with trimethoprim and sulfamethoxazole, and remember that probiotics carry a small risk of bloodstream infection in fragile infants with central lines. Above all, prophylaxis never replaces a blood culture and a full intravenous course when the child is febrile or her stools turn pale.

Part six. Bile lakes.

Some children develop cystic dilatations of the bile ducts inside the liver after the Kasai. These are called intrahepatic biliary cysts or bile lakes, and they are reported in between about six and twenty-six percent of children, with higher figures in series that image routinely and follow children for longer. In a Taiwanese series of one hundred and fifty-four children published in 2002, a quarter had at least one cyst and about one in ten had multiple cysts. Among those with multiple cysts, almost ninety-four percent had already had cholangitis, and mortality was higher than in children with a single cyst. In a Chinese series of seven hundred and eighty-seven children, cystic lesions were found in about ten percent, seventy percent of whom had cholangitis. Two-year native-liver survival was about sixty percent in children with cysts and cholangitis, against eighty-seven percent in those with cysts but no cholangitis. A Japanese group in 2024 found bile lakes in twenty-seven of sixty-eight children who still had their own liver at one year, and a score based on the bile lakes, combined with the serum bile acid level, predicted later transplant with an area under the curve of about zero point eight three.

It helps to think of two kinds of lesion. A single simple lesion, usually near the porta, is often a pseudocyst. Histology shows that it has no biliary lining; it is a collection of bile that has escaped from the ductules. Such a lesion can often be drained. Multiple or complex dilatations of true ducts throughout the liver are different. They reflect diffuse obstruction and fibrosis, they tend to recur, and they carry a worse prognosis. A biloma is a third thing: a collection of

bile outside the ducts after a leak. It is drained like any other collection, and there are no series specific to biliary atresia.

Management is stepwise and rests on small series. First, recognise the lakes, with an ultrasound at every episode of cholangitis and at routine follow-up. Second, give a prolonged intravenous course of antibiotics, because the lake is a closed, infected space that the blood level of antibiotic reaches poorly. Third, for a single lake, consider percutaneous drainage. In a 2024 series, drainage resolved the lake in six of seven children, but three of the seven developed biliary peritonitis, so the procedure is not trivial. Fourth, consider a surgical bile-lake jejunostomy, in which the lake is drained into a loop of bowel. In a Finnish series published in 2019, this was done in six children, the lakes regressed in all of them, and the rate of cholangitis fell from almost nine episodes a year to about one. The same series reported that two children who had earlier been treated with percutaneous drainage alone died. In a small Chilean series, the time free of cholangitis was only about a month and a half in children with bile lakes, against more than two years in those without, and drainage worked only for single cysts. Finally, children with multiple lakes and recurrent cholangitis, or with failing drainage, should be assessed for transplant. Because the series are so small, these decisions are individual and belong in a centre that offers both hepatobiliary surgery and liver transplantation.

Part seven. What the guidelines say.

It is striking that the commonest complication of the Kasai has no society guideline. Neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has a treatment guideline for biliary atresia. The 2017 guideline of the North American and European societies for paediatric gastroenterology, hepatology and nutrition, known as NASPGHAN and ESPGHAN, is about the diagnosis of cholestasis in infants and does not address cholangitis after the Kasai. The Japanese Biliary Atresia Society guideline of 2021 makes no recommendation on steroids, and a second edition appeared in 2026. In practice, then, the BARD consensus of 2022 is the standard. On the definition, on the choice of empiric antibiotic and on bile lakes, there is simply a gap: no society document speaks. On duration, the consensus recommends ten to twenty-one days, but practice is often shorter, so practice differs from consensus. On primary prophylaxis there is a material conflict: the consensus supports six to twelve months of prophylaxis, while the meta-analyses of 2023 and 2026 show no reduction in cholangitis. On steroids and other adjuncts, the documents that speak agree that there is not enough evidence to recommend them.

Part eight. Traps and pearls.

Several errors in reasoning recur. The first is to say that a child without fever cannot have cholangitis. Fever was present in only about four in ten episodes in the validation study, while pale stools and a rising GGT were far more common. The second is to call a culture-negative

episode viral. Cultures are positive in only ten to forty-four percent of episodes, so you must treat on the clinical and laboratory criteria. The third is to declare that the Kasai has failed when the bilirubin rises during a fever. Treat the infection fully first, because the bilirubin may fall again. The fourth is to dismiss cholangitis because the child is on prophylaxis. Breakthrough infection is common, and the right response is to change to a different class of antibiotic. The fifth is to switch to oral antibiotics after three days because the child looks well. The consensus advises ten to twenty-one days, and short courses relapse. And one more: a habit of always giving ceftriaxone, in a unit where Klebsiella and enterococci dominate, means that many first courses will fail. Know the local antibiogram and the child's last isolate.

Let us return to the scenarios that come up in clinic. An infant four weeks after the Kasai who is afebrile but has pale stools and a GGT that has doubled should have blood cultures and start intravenous antibiotics as suspected cholangitis, because fever is absent in most episodes. An infant with fever at eight weeks while on trimethoprim and sulfamethoxazole should receive meropenem or piperacillin with tazobactam, because breakthrough on prophylaxis selects resistance. If the culture grows enterococcus faecium, add vancomycin and narrow later, because only about four in ten of these isolates are susceptible to ampicillin. If the child is well on day five and the parents ask for oral treatment, complete ten to fourteen days intravenously. A third episode within six months calls for an ultrasound to look for bile lakes and a referral to the transplant team, because relapsing cholangitis predicts loss of the liver. A single lake near the porta with recurrent fever calls for prolonged antibiotics and a discussion about drainage, since single lakes drain and multiple lakes usually do not. And when a family asks whether prophylaxis is really needed, it is honest to offer six to twelve months while explaining that the benefit is not proven.

So, tomorrow in clinic, carry these points. Half or more of infants have cholangitis after the Kasai, most in the first year and often by two months of age. Partial drainage carries the highest risk, but good early drainage still brings frequent cholangitis. The BARD definition calls cholangitis suspected with one clinical and one laboratory criterion, and confirmed with two of each or a positive culture. Fever was present in only about forty-one percent of episodes, whereas bilirubin or GGT rose in ninety percent. Blood cultures are positive in only ten to forty-four percent of episodes, and enterococci, Escherichia coli, Enterobacter and Klebsiella are the usual organisms. Third-generation cephalosporins are failing more often, so keep meropenem for severe or resistant cases. Treat for ten to fourteen days when cholangitis is suspected and fourteen to twenty-one days when it is confirmed, and escalate if there is no response at seventy-two hours. Immunoglobulin shortened fever in severe cholangitis in one trial, but it is not routine. Prophylaxis for six to twelve months is consensus, and the meta-analyses show no fall in cholangitis. And bile lakes, found in six to twenty-six percent of children, are a warning sign: multiple lakes with recurrent cholangitis point towards transplant. The operation creates an open road from the bowel to the liver. Our job afterwards is to notice

early when bacteria travel along it, to treat them fully, and to recognise when the road has done more harm than good.